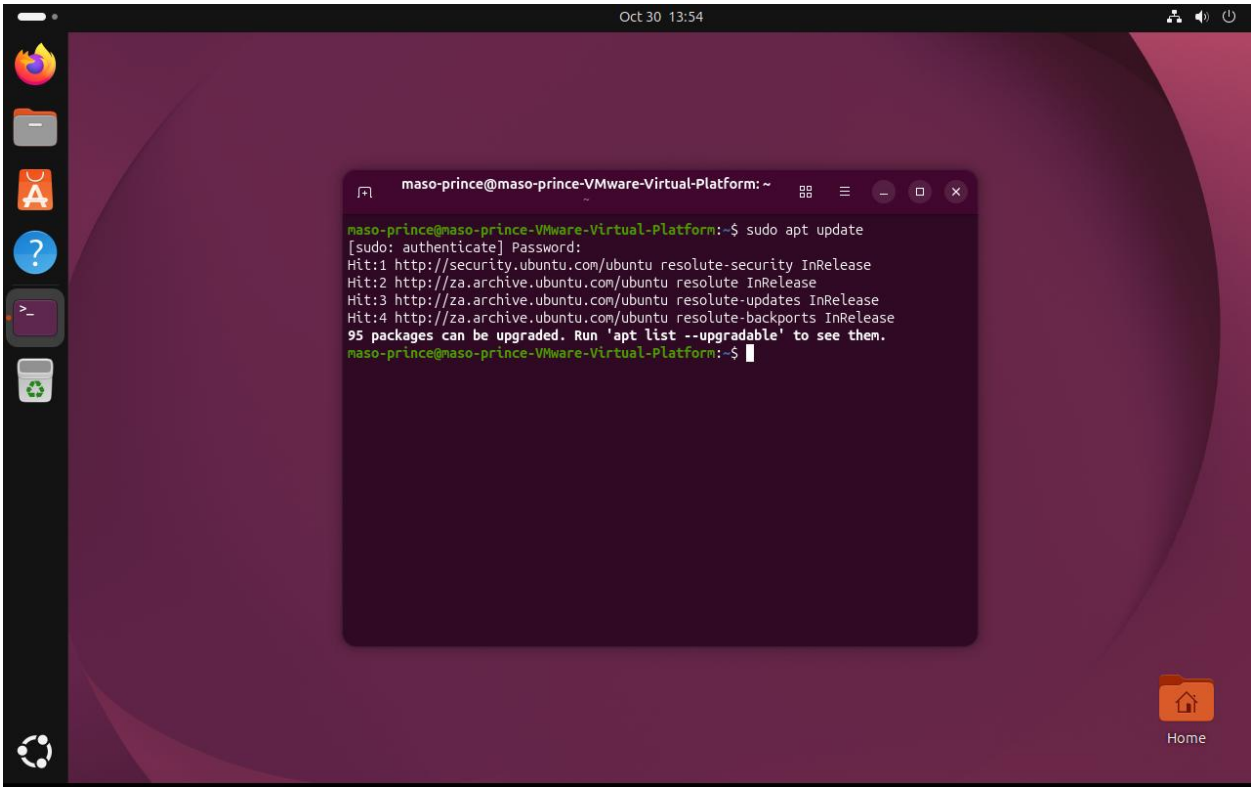
NET732

Week 1: CyberOps & Lab Setup

Practical

1. Lab setup

I installed VMware Workstation Pro on my Windows 11 computer and created an Ubuntu 26.04 virtual machine. I allocated 4 GB RAM, 2 processor cores and 30 GB storage. I configured the network to use NAT so Ubuntu could access the internet.



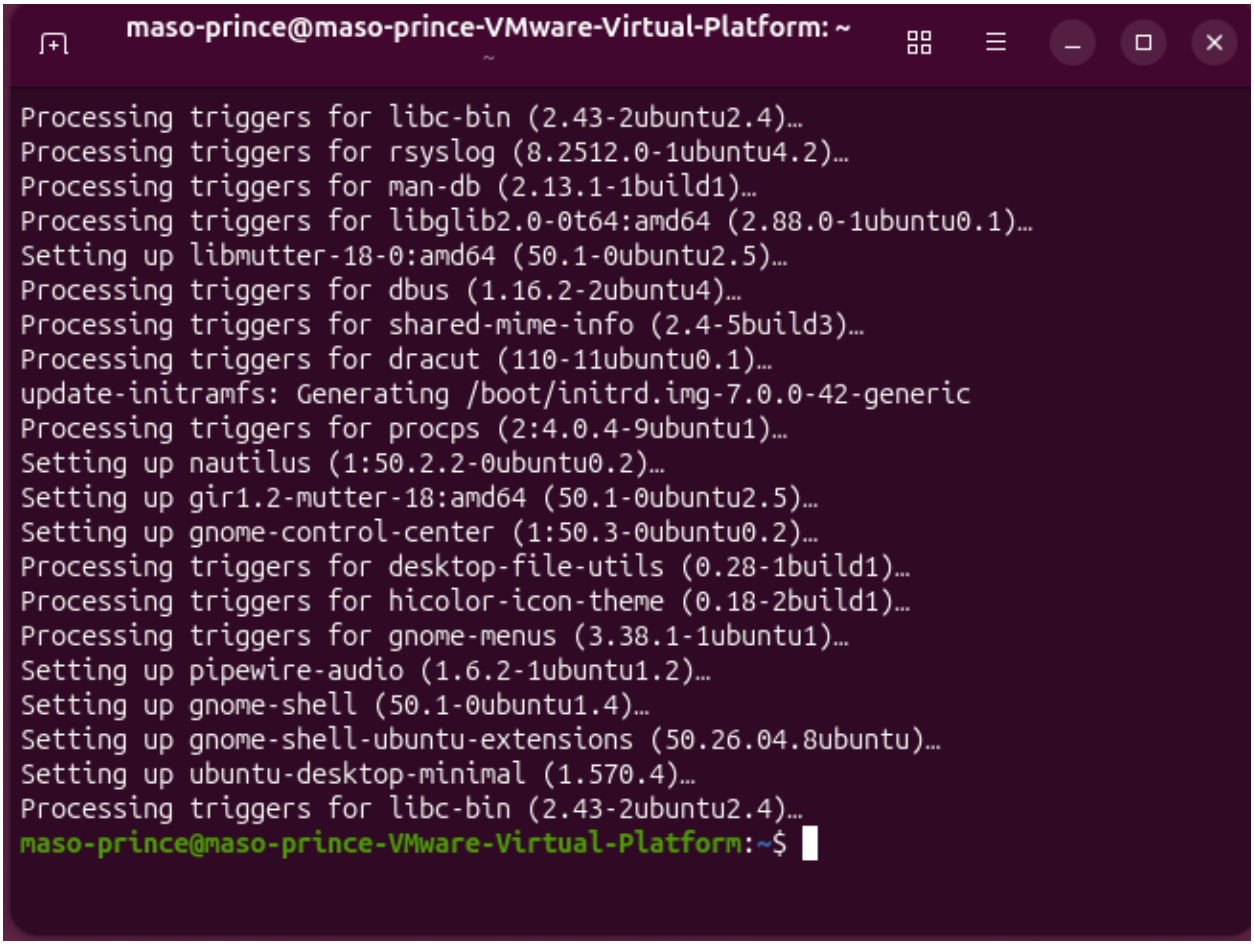
I used VMware Workstation Pro as an alternative to VirtualBox.

2. Updating Ubuntu

I updated my Ubuntu system using:

sudo apt update

sudo apt upgrade -y



These commands check for and install available system updates.

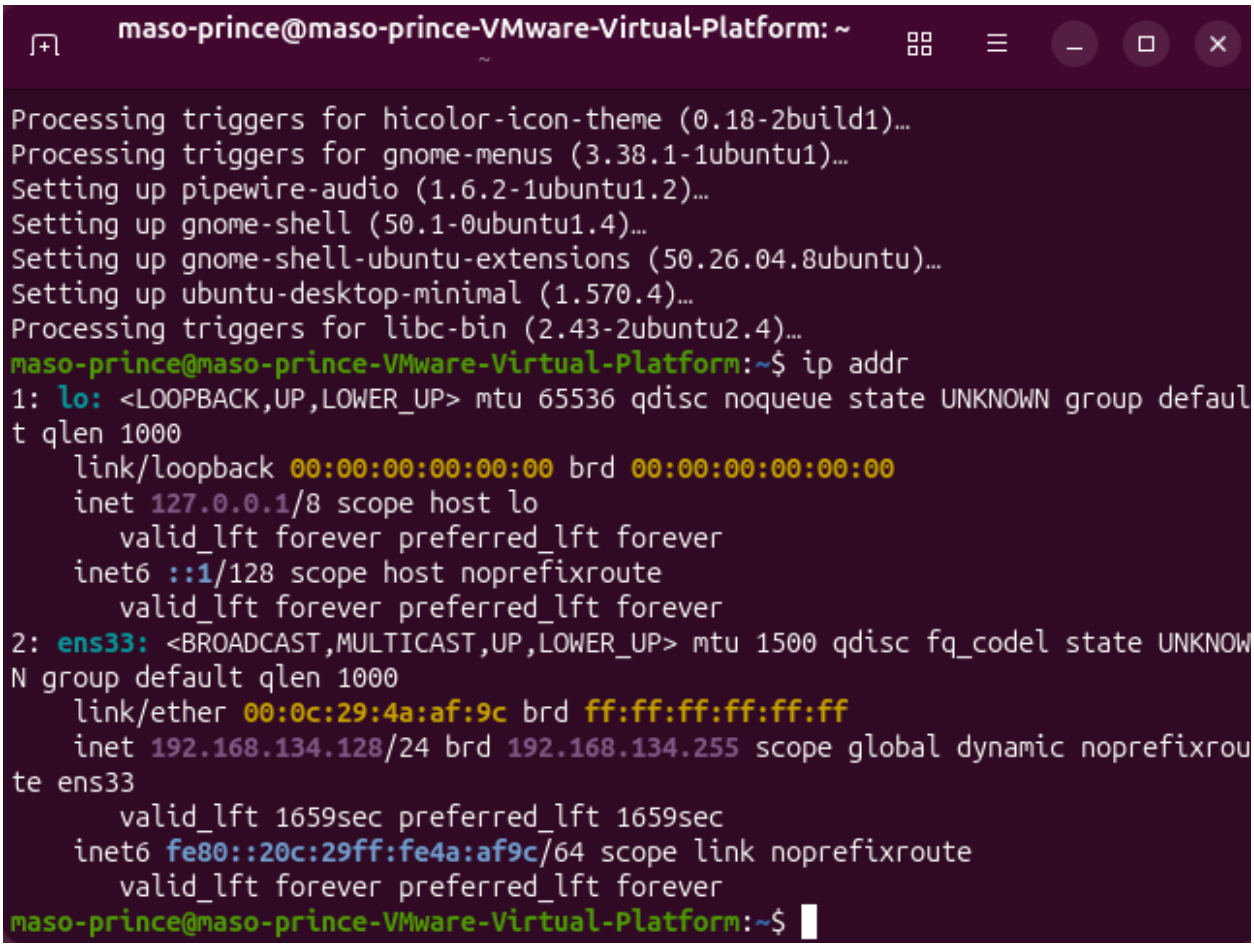
3. Checking network configuration

I used the following commands:

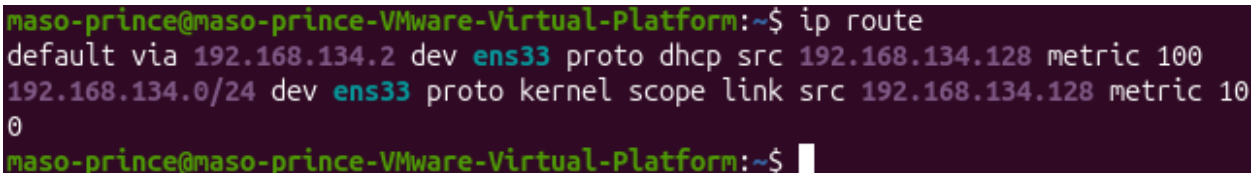
ip addr

ip route

I used ip addr to check my network interfaces and IP addresses.



I used ip route to view my network routes and default gateway.



4. Testing connectivity

I tested my internet connection using:

ping -c 4 8.8.8.8

ping -c 4 google.com

```
maso-prince@maso-prince-VMware-Virtual-Platform:~$ ping -c 4 google.com\
>
PING google.com (142.251.216.78) 56(84) bytes of data.
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=1 ttl=128 ti
me=11.5 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=2 ttl=128 ti
me=8.22 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=3 ttl=128 ti
me=30.8 ms
64 bytes from lcjnbi-ae-in-f14.1e100.net (142.251.216.78): icmp_seq=4 ttl=128 ti
me=9.63 ms

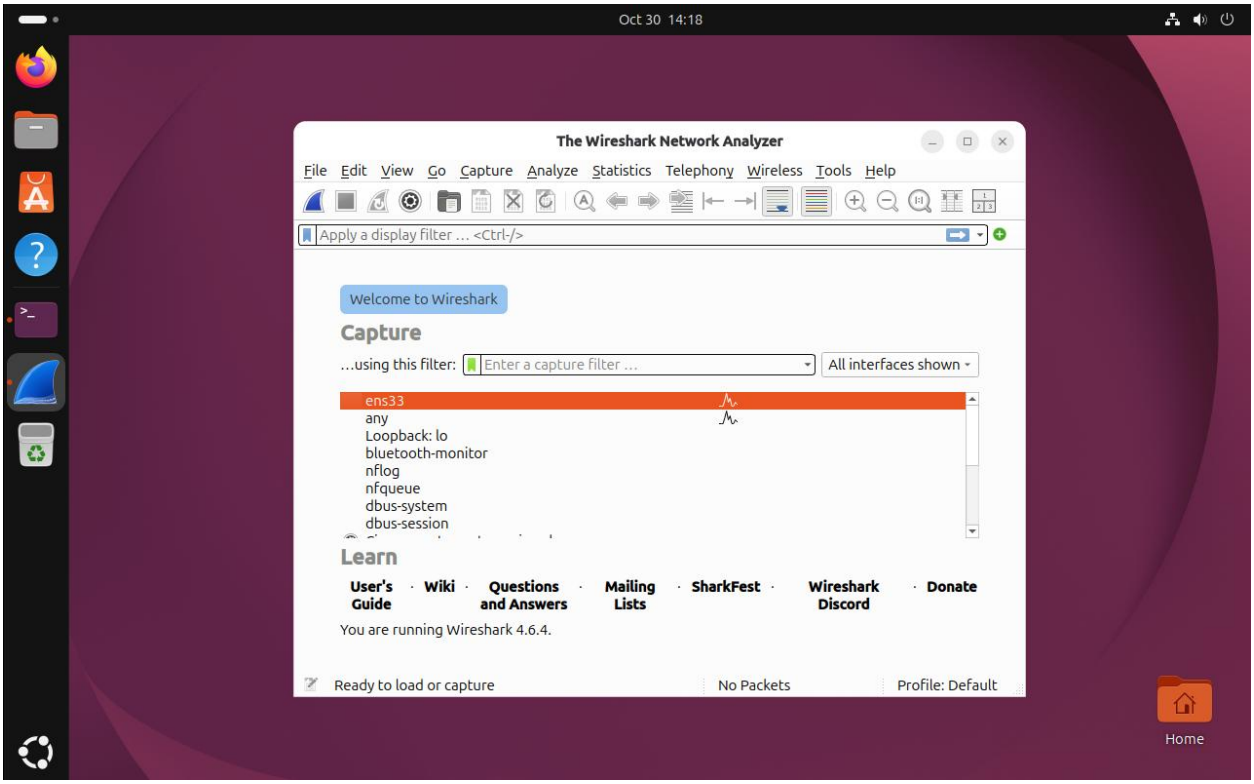
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 8.220/15.026/30.801/9.179 ms
maso-prince@maso-prince-VMware-Virtual-Platform:~$
```

These commands helped me check internet connectivity and DNS resolution.

5. Installing and using Wireshark

I installed Wireshark using:

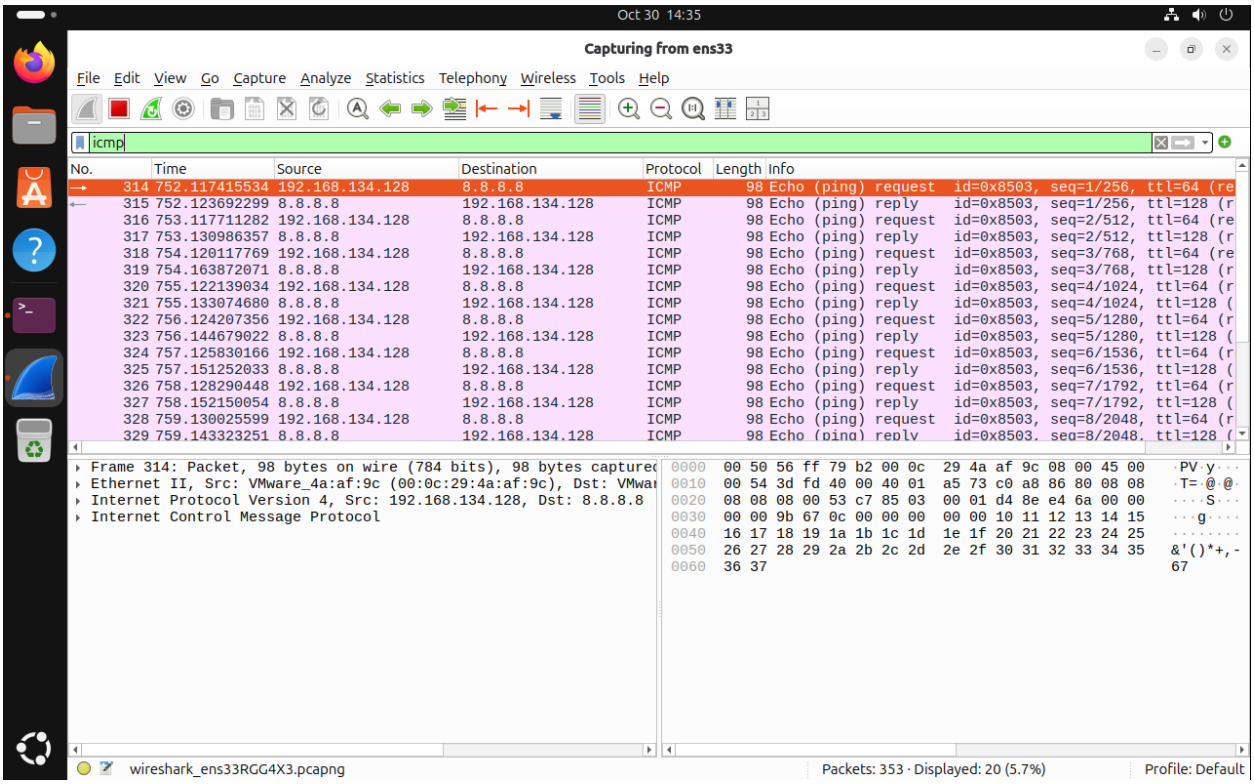
sudo apt install wireshark -y



I opened Wireshark, selected my active network interface and started capturing packets. I then generated network traffic using:

ping -4 -c 10 8.8.8.8

I applied the icmp filter in Wireshark to view ICMP packets. I observed Echo Requests and Echo Replies.



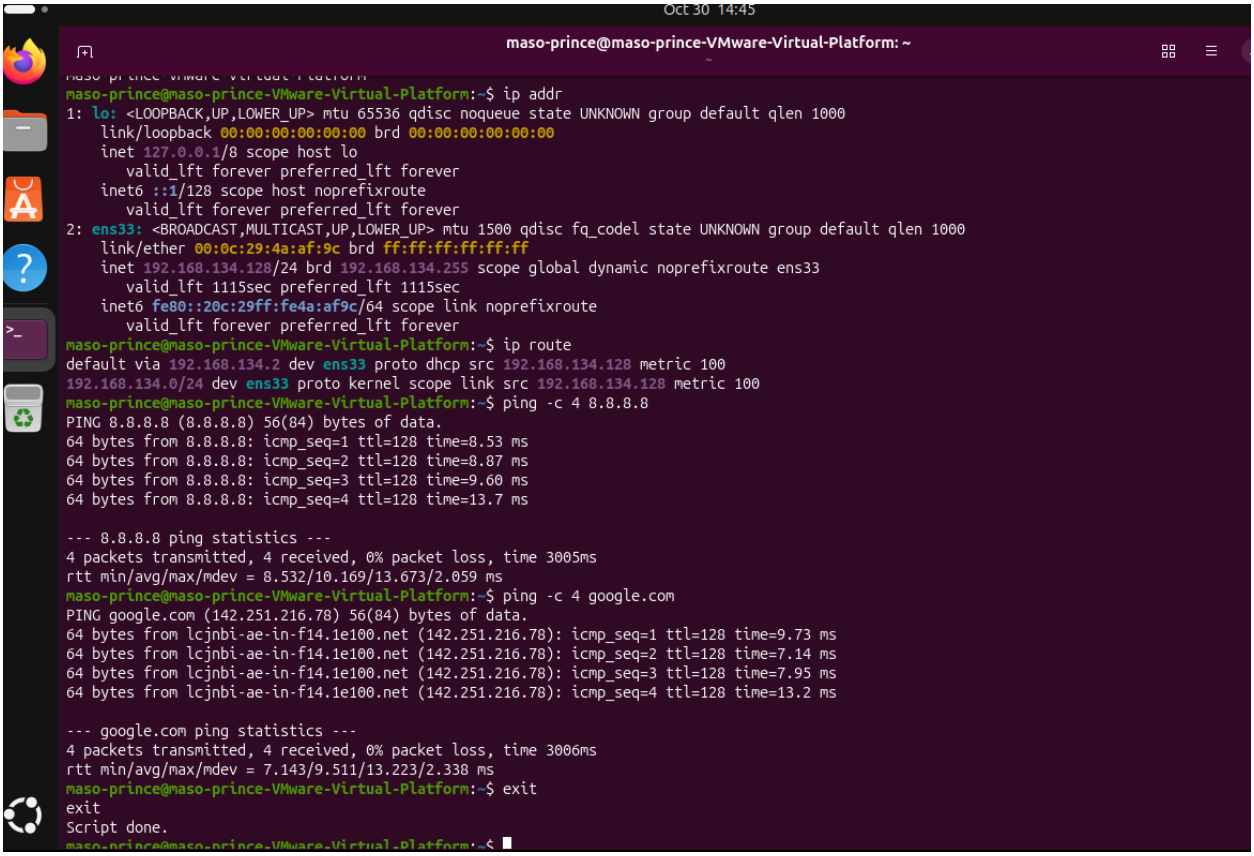
6. Saving evidence and logs

I saved my packet capture as:

CyberOps_Week1_Capture.pcapng

I also recorded my terminal commands and outputs using:

script -a ~/CyberOps/Week1/commands.log



7. Reflection and conclusion

I observed how network devices communicate through IP addresses and ICMP packets. I used Wireshark to capture and analyse network traffic. This relates to SOC operations because security analysts monitor network traffic to identify suspicious activities and collect evidence.

I successfully established my CyberOps laboratory and tested network connectivity. My next step is to practise analysing DNS and TCP traffic to improve my network monitoring skills.